

Limiting Reactants and Percent Yield

Answer key with worked solutions

A. Understanding

1. Definitions.

(a) The reactant used up first, which limits the amount of product. (b) A reactant left over when the reaction stops. (c) The maximum amount of product, calculated from the limiting reactant.

2. Why yields are below 100%.

Any two of: the reaction does not go to completion; side reactions form other products; product is lost during separation or purification.

B. Working in moles

3. 3.0 mol H₂ + 2.0 mol O₂.

$3.0 \text{ mol H}_2 \times (1 \text{ mol O}_2 / 2 \text{ mol H}_2) = 1.5 \text{ mol O}_2$ needed, so (a) **H₂ is limiting**. (b) $\text{H}_2\text{O} = 3.0 \text{ mol H}_2 \times (2 \text{ mol H}_2\text{O} / 2 \text{ mol H}_2) = \mathbf{3.0 \text{ mol}}$. (c) $\text{O}_2 \text{ left} = 2.0 \text{ mol} - 1.5 \text{ mol} = \mathbf{0.5 \text{ mol}}$.

4. 1.5 mol N₂ + 3.0 mol H₂.

$1.5 \text{ mol N}_2 \times (3 \text{ mol H}_2 / 1 \text{ mol N}_2) = 4.5 \text{ mol H}_2$ needed, but only 3.0 mol is available, so (a) **H₂ is limiting**. (b) $\text{NH}_3 = 3.0 \text{ mol H}_2 \times (2 \text{ mol NH}_3 / 3 \text{ mol H}_2) = \mathbf{2.0 \text{ mol}}$. (c) $\text{N}_2 \text{ used} = 3.0 \text{ mol H}_2 \times (1 \text{ mol N}_2 / 3 \text{ mol H}_2) = 1.0 \text{ mol}$, so $\text{N}_2 \text{ left} = 1.5 \text{ mol} - 1.0 \text{ mol} = \mathbf{0.5 \text{ mol}}$.

C. Working in grams

5. 5.40 g Al + 15.0 g Cl₂.

$n(\text{Al}) = 5.40 \text{ g} \div 26.98 \text{ g/mol} = 0.2001 \text{ mol}$; $n(\text{Cl}_2) = 15.0 \text{ g} \div 70.90 \text{ g/mol} = 0.2116 \text{ mol}$.

The Al would need $0.2001 \text{ mol Al} \times (3 \text{ mol Cl}_2 / 2 \text{ mol Al}) = 0.3002 \text{ mol Cl}_2$, but only 0.2116 mol is available, so (a) **Cl₂ is limiting**.

(b) $n(\text{AlCl}_3) = 0.2116 \text{ mol Cl}_2 \times (2 \text{ mol AlCl}_3 / 3 \text{ mol Cl}_2) = 0.1410 \text{ mol}$; $M(\text{AlCl}_3) = 133.33 \text{ g/mol}$; mass = $0.1410 \text{ mol} \times 133.33 \text{ g/mol} = \mathbf{18.8 \text{ g}}$.

(c) Al used = 0.1410 mol; Al left = $0.2001 \text{ mol} - 0.1410 \text{ mol} = 0.0591 \text{ mol}$; $0.0591 \text{ mol} \times 26.98 \text{ g/mol} = \mathbf{1.59 \text{ g}}$.

6. Percent yield (15.2 g collected).

Percent yield = $15.2 \text{ g} \div 18.81 \text{ g} \times 100\% = \mathbf{80.8\%}$ (using the unrounded theoretical yield, 18.81 g; the units of g cancel).

7. 10.0 g CH₄ + 32.0 g O₂.

$n(\text{CH}_4) = 10.0 \text{ g} \div 16.04 \text{ g/mol} = 0.6234 \text{ mol}$; $n(\text{O}_2) = 32.0 \text{ g} \div 32.00 \text{ g/mol} = 1.000 \text{ mol}$. The methane needs $0.6234 \text{ mol CH}_4 \times (2 \text{ mol O}_2 / 1 \text{ mol CH}_4) = 1.247 \text{ mol O}_2$, so (a) **O₂ is limiting**.

(b) $n(\text{CO}_2) = 1.000 \text{ mol O}_2 \times (1 \text{ mol CO}_2 / 2 \text{ mol O}_2) = 0.500 \text{ mol}$; mass = $0.500 \text{ mol} \times 44.01 \text{ g/mol} = \mathbf{22.0 \text{ g}}$.

(c) CH₄ used = $0.500 \text{ mol} \times 16.04 \text{ g/mol} = 8.02 \text{ g}$; CH₄ left = $10.0 \text{ g} - 8.02 \text{ g} = \mathbf{1.98 \text{ g}}$.

D. Challenge

8. A 108% yield.

Most likely the product was still **wet** or contained **impurities**, which added mass. Mass cannot be created, so a pure, dry product can never exceed the theoretical yield.